Exploratory Data Analysis Report

Univariate Analysis

Dataset Name

https://www.kaggle.com/datasets/satyajeetrai/palmer-penguins-dataset-for-eda

Analysis Code

https://colab.research.google.com/drive/1JGWzkmzex85tIklt3VHrRJxZAQnkEGx5?usp=sharing

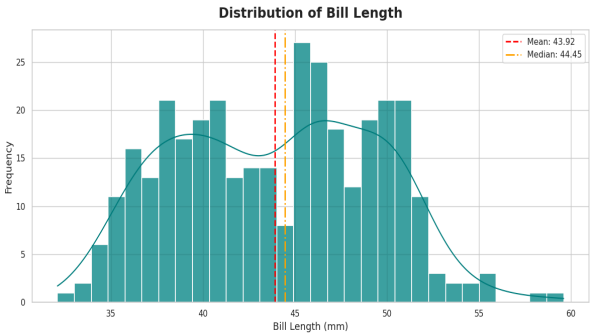
Describe the Columns

Analysis	Spacies	Island	Sex
Count	344	344	333
Unique	3	3	2
Top	Adelie	Biscoe	male
Frequency	152	168	168
Data type	object	object	object

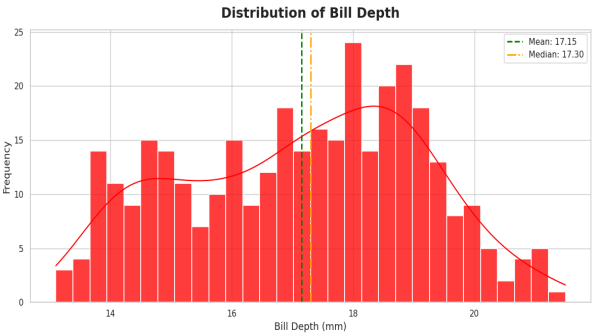
Analysis	bill_length	bill_depth	flipper_length	body_mass
Count	342.000000	342.000000	342.000000	342.000000
Mean	43.921930	17.151170	200.915205	4201.754386
Standard Deviation	5.459584	1.974793	14.061714	801.954536
Minimum	32.100000	13.100000	172.000000	2700.000000
Less than 25% data	39.225000	15.600000	190.000000	3550.000000
Less than 50% data	44.450000	17.300000	197.000000	4050.000000
Less than 75% data	48.500000	18.700000	213.000000	4750.000000
Maximum	59.600000	21.500000	231.000000	6300.000000

Data Distributions

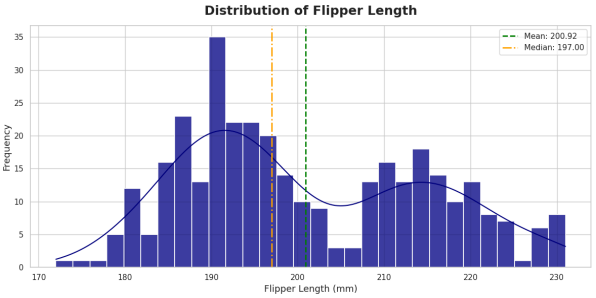
Skewness value	Interpretation	Skewness value	Interpretation
Approximately 0	Symmetric distribution	−0.5 to 0	Slight left skew
0 to +0.5	Slight right skew	−1 to −0.5	Moderate left skew
+0.5 to +1	Moderate right skew	Less than −1	Strong left skew
Greater than +1	Strong right skew		



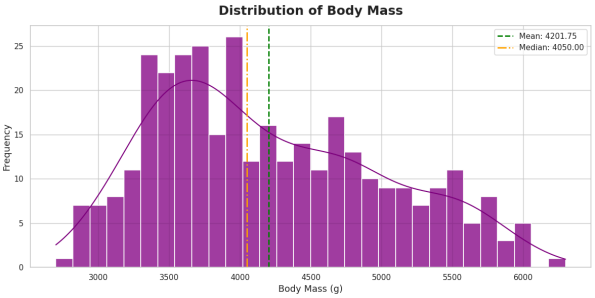
Skewness: 0.05311806699132413



Skewness: -0.14346462519436454



Skewness: 0.3456818328687695

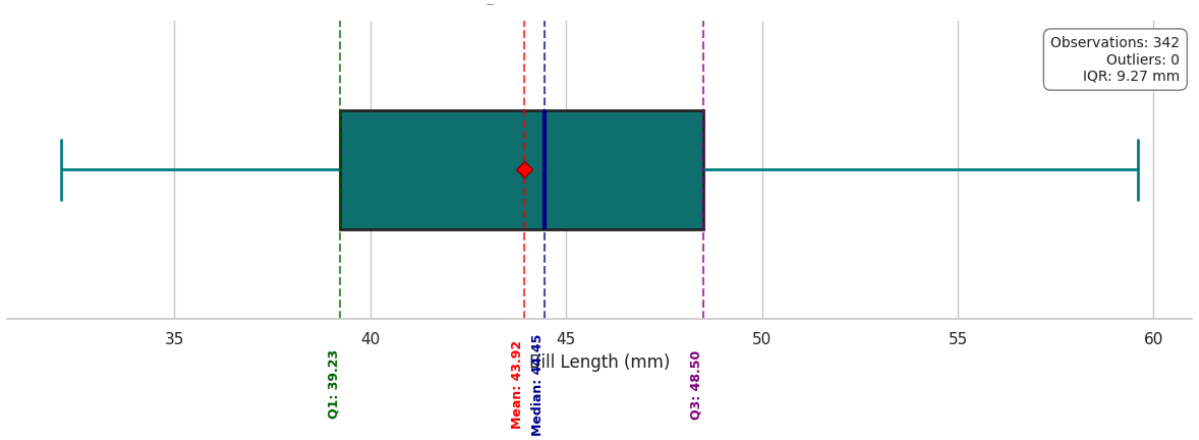


Skewness: 0.4703293304801231

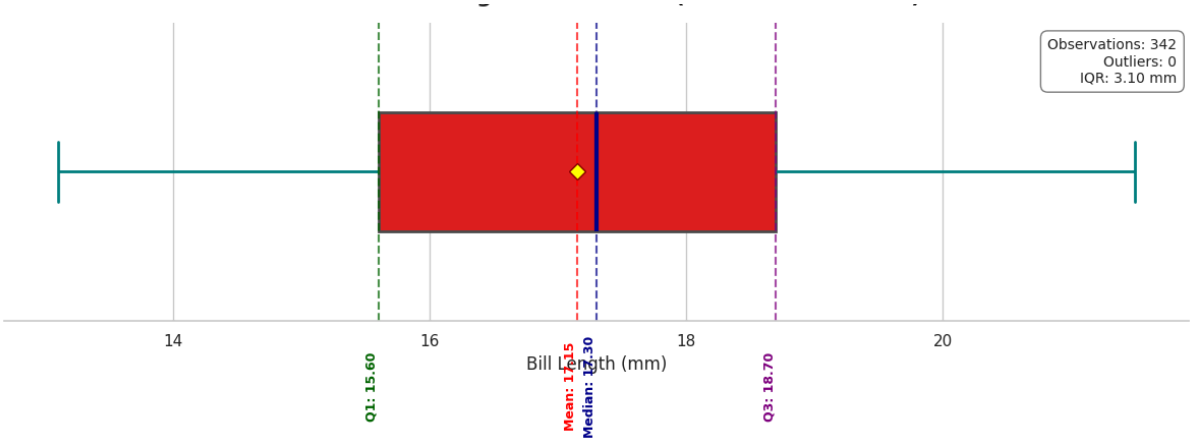
Log Transformations Based on the Skewness Value

Skewness	Interpretation	Log Transform
−0.5 to +0.5	Approximately symmetric	Usually unnecessary
+0.5 to +1.0	Moderately right-skewed	Consider it
Greater than +1.0	Strongly right-skewed	Often worth testing
Less than -1.0	Strongly left-skewed	Log is not the first choice

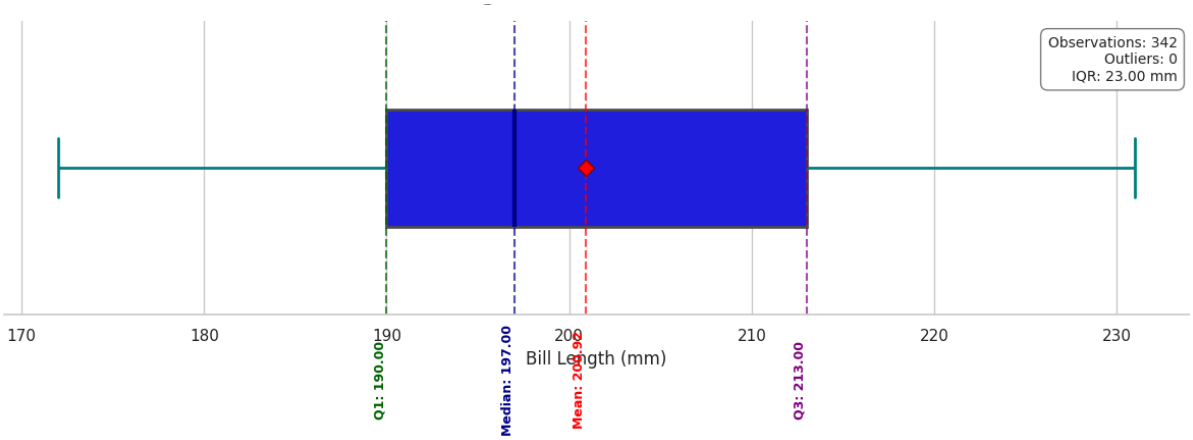
Outliers Detection and IQR



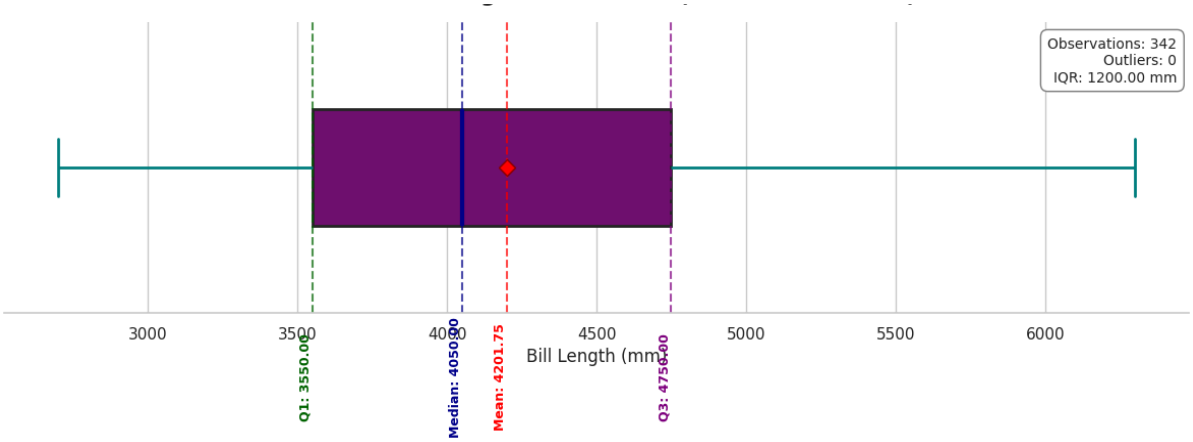
Box Plot of Bill Length



Box Plot of Bill Depth



Box Plot of Flipper Length



Box Plot of Body Mass

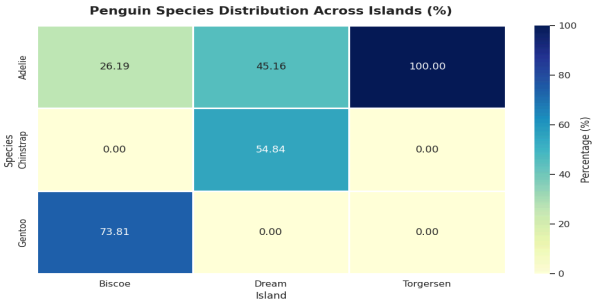
NULL Values

Columns	Sex	Bill Length	Bill Depth	Flipper Length	Body Mass
NULL values	11	2	2	2	2

Bivariate Analysis

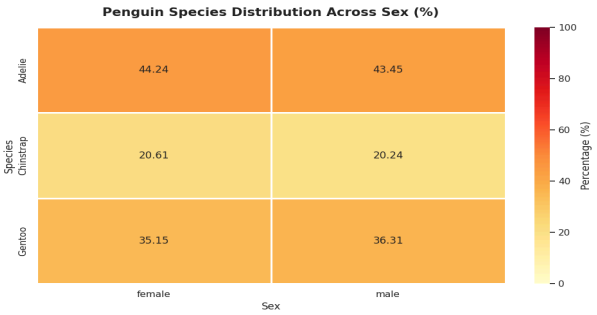
Species VS Island Crosstab

	Biscoe	Dream	Torgersen
Adelie	26.19%	45.16%	100%
Chinstrap	0.00%	54.83%	0.00%
Gentoo	73.80%	0.00%	0.00%



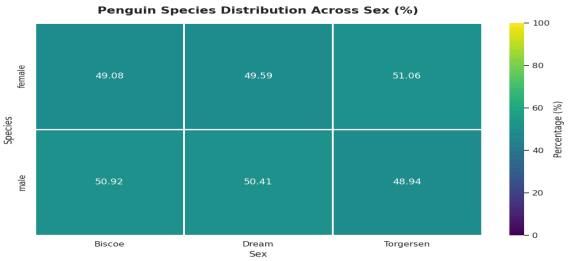
Species VS Sex Crosstab

	Female	Male
Adelie	44.24%	43.45%
Chinstrap	20.60%	20.23%
Gentoo	35.15%	36.30%



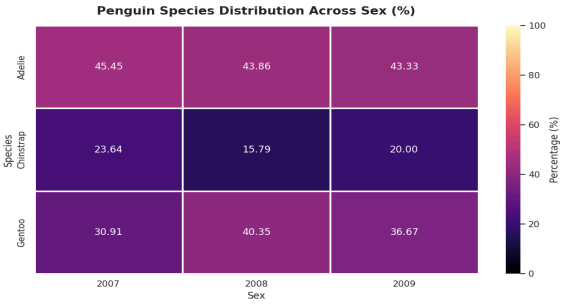
Sex VS Island Crosstab

	Biscoe	Dream	Torgersen
Female	49.07%	49.59%	51.06%
Male	50.92%	50.40%	48.93%



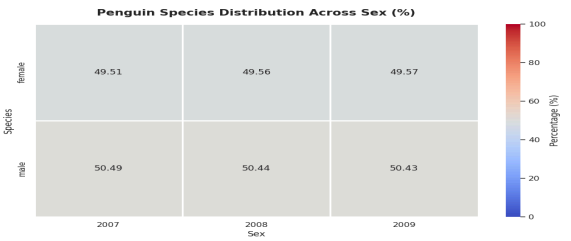
Species VS Year Crosstab

	2007	2008	2009
Adelie	45.45%	43.85%	43.33%
Chinstrap	23.63%	15.78%	20.00%
Gentoo	30.90%	40.35%	36.66%



Sex VS Year Crosstab

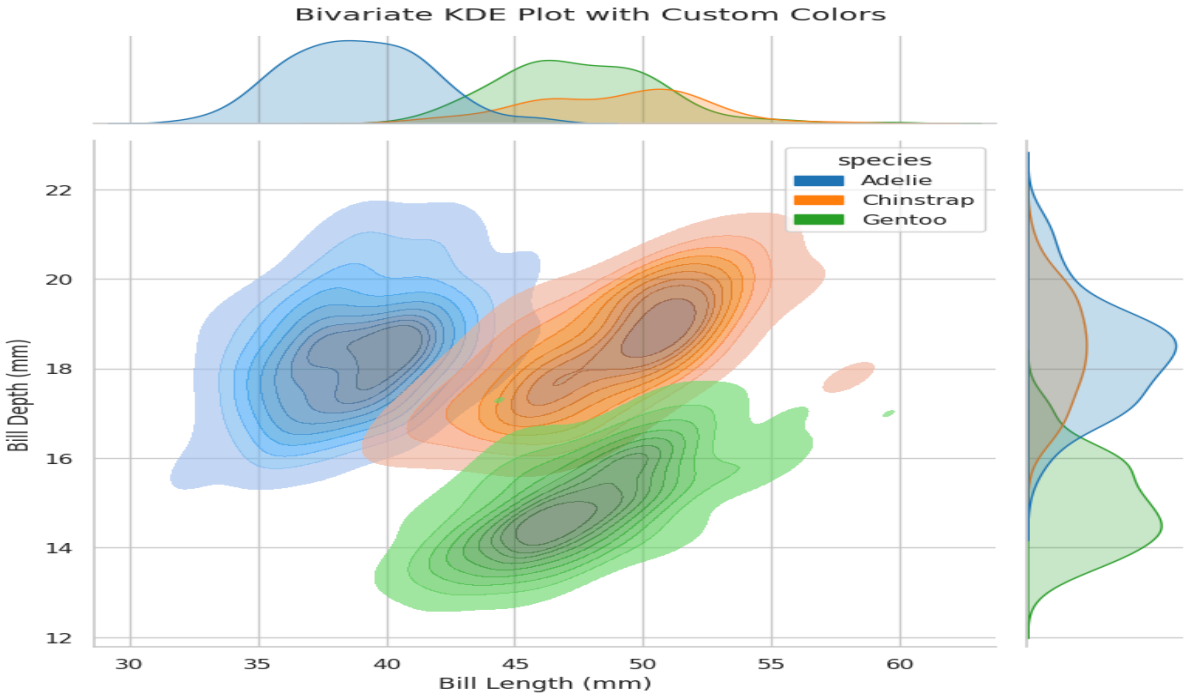
	2007	2008	2009
Female	49.51%	49.55%	49.57%
Male	50.48%	50.44%	50.42%



Dataset Shape

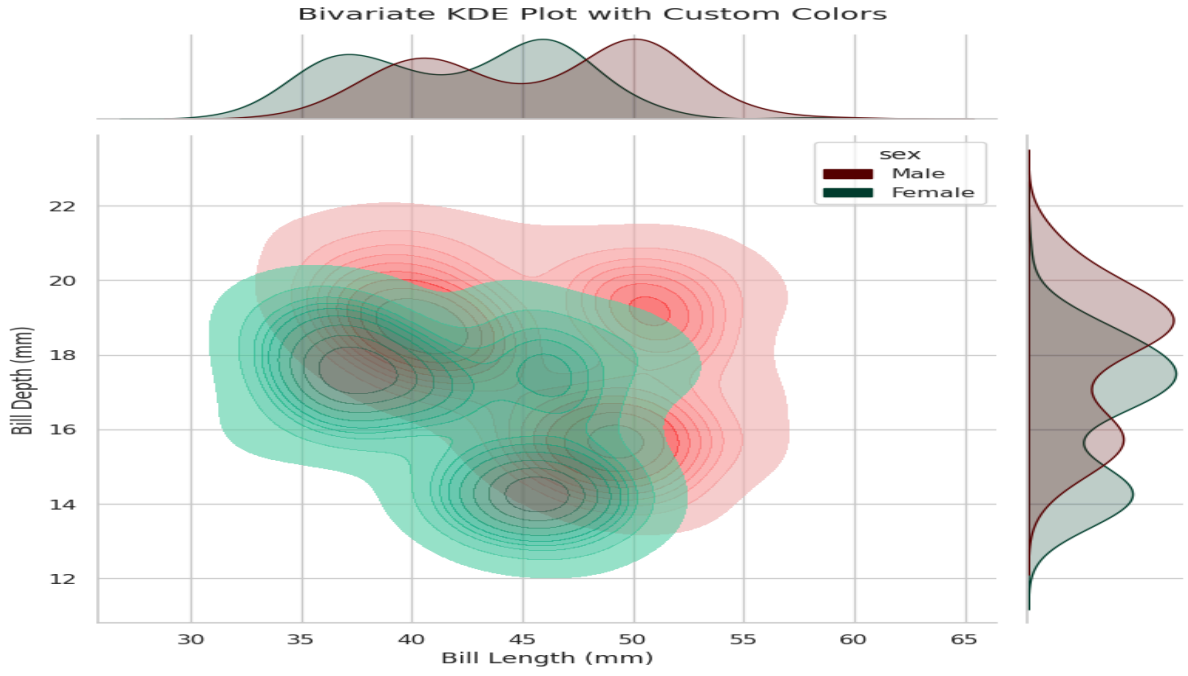
Total Rows	344 rows
Total Columns	9 features

KDE analysis between bill_length and bill_depth based on the species



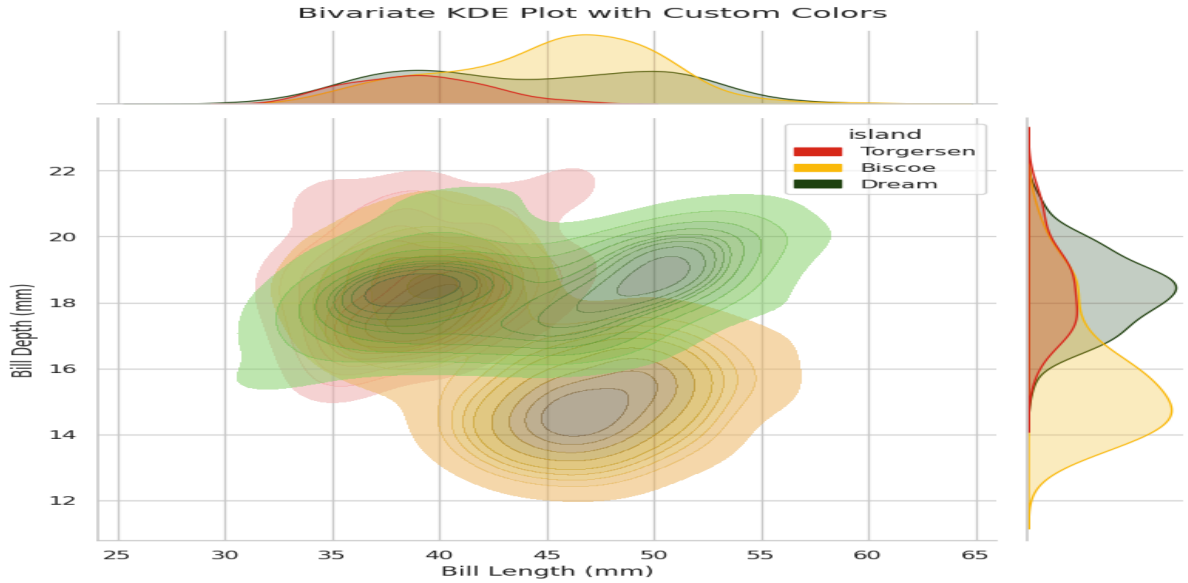
Most of the Adelie has a bill length between 36mm to 42mm with a bill depth of 17mm to 19mm. Most of the Chinstrap has a bill length between 48mm to 53mm with a bill depth of 18mm to 20mm. Most of the Gentoo has a bill length between 45mm to 49mm with a bill depth of 14mm to 15mm. Bill length between 39mm to 45mm with a bill depth between 16mm to 19mm can be classified both Adelie and Chinstrap simultaneously. Bill length between 47mm to 54mm with a bill depth between 15.6mm to 17.8mm can be classified both Gentoo and Chinstrap simultaneously. If the bill length is decreased, the bill depth is also decreased. And if the bill length is increased, the bill depth is also increased.

KDE analysis between bill_length and bill_depth based on the sex



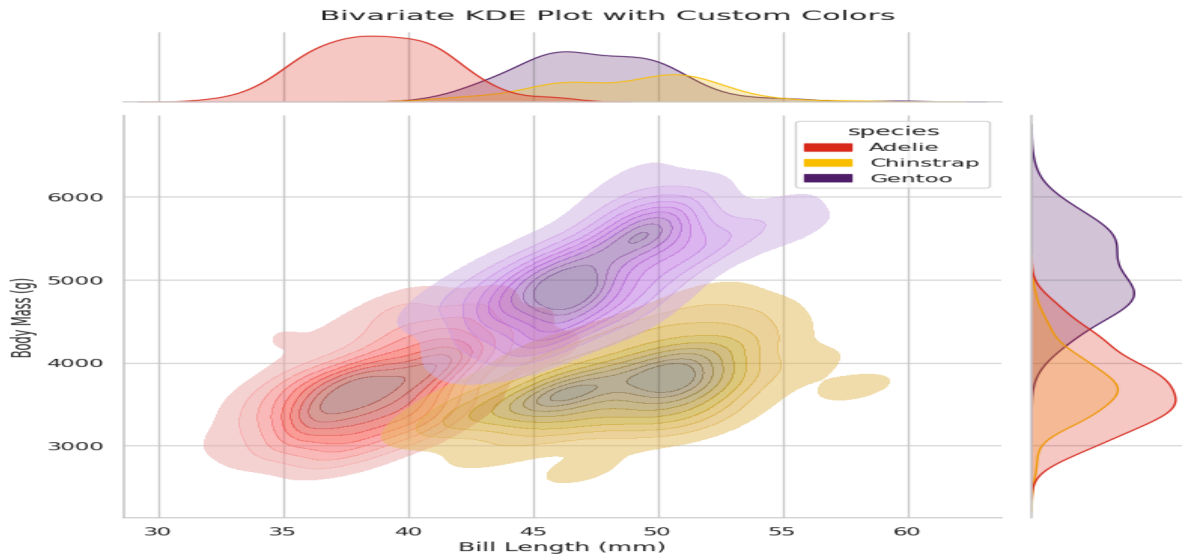
Most of the females have a bill length between 35mm to 42mm with a bill depth of 16.5mm to 19.5mm. And a bill length between 44mm to 47mm with a bill depth of 14mm to 15mm. Most of the Male have a bill length between 48mm to 54mm with a bill depth of 15mm to 17mm. And a bill length between 37mm to 41mm with a bill depth of 18mm to 20mm. Bill length between 33mm to 51mm with a bill depth between 13.6mm to 19.5mm can be classified both Female and Male simultaneously. Bill length between 46mm to 52mm with a bill depth between 15.5mm to 17.5mm can be classified both Female and Male simultaneously. If the bill length is decreased, the bill depth is increased. And if the bill length is increased, the bill depth is also decreased.

KDE analysis between bill_length and bill_depth based on the island



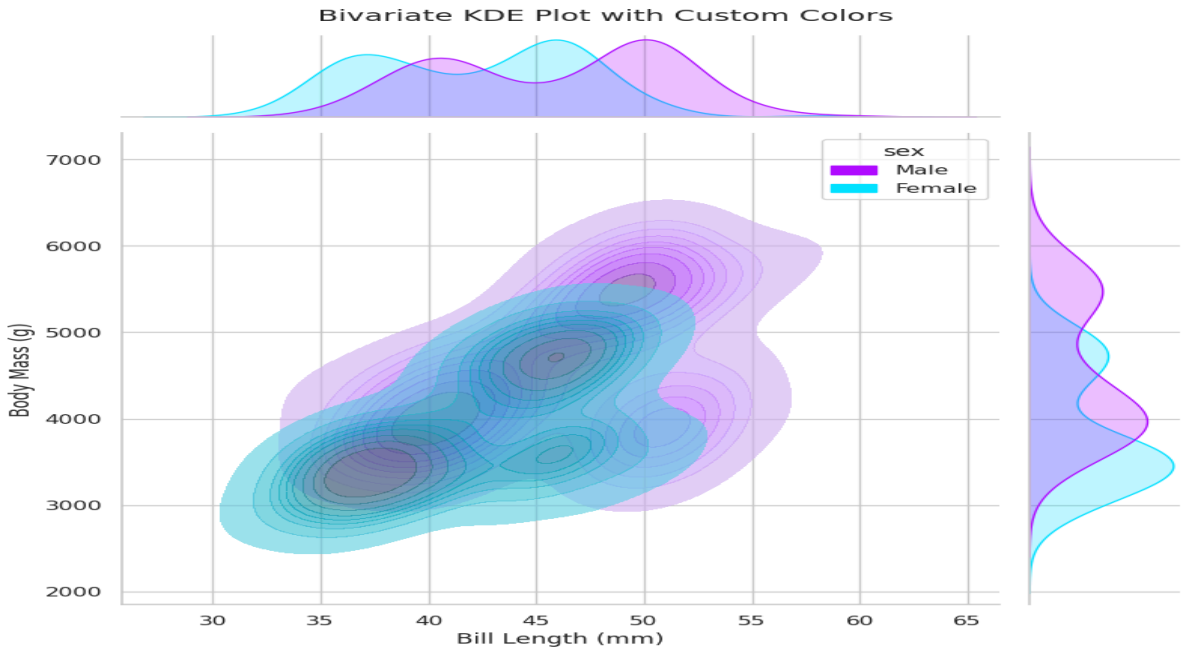
The penguins in Torgersen, Biscoe and Dream have the common bill length range of 34.6mm to 45mm with a bill depth of 16mm to 17mm. However, in Biscoe, the bill length increases while the bill depth decreases. But on the other hand, the penguins living in Dream island have a large bill length with deep bill depth. Penguins living in Torgersen have deep bill depth but small bill length. Biscoe have the lowest bill depth penguins.

KDE analysis between bill_length and body_mass based on the species



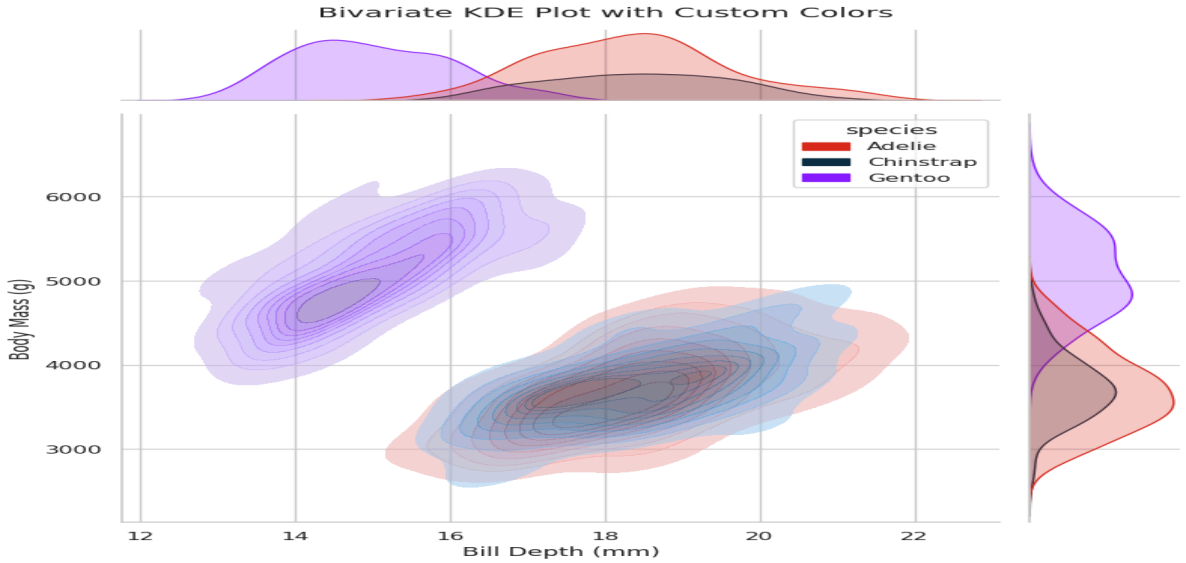
Among all the species, Adelie have the lowest bill length and body mass. And for all the species, as the bill length increases the body mass also increases. Species who have the bill length range between 42mm to 46mm have the same body mass around 4KG. However, Gentoo has the highest body mass of around 6KG. Most of the Adelie has a bill length of 36mm to 39mm with a body mass of 3500 grams. Most of the Chinstrap has a bill length of 48mm to 51mm with a body mass of 3800 grams. Most of the Adelie has a bill length of 46mm to 48mm with a body mass of 4900 grams.

KDE analysis between bill_length and body_mass based on the sex



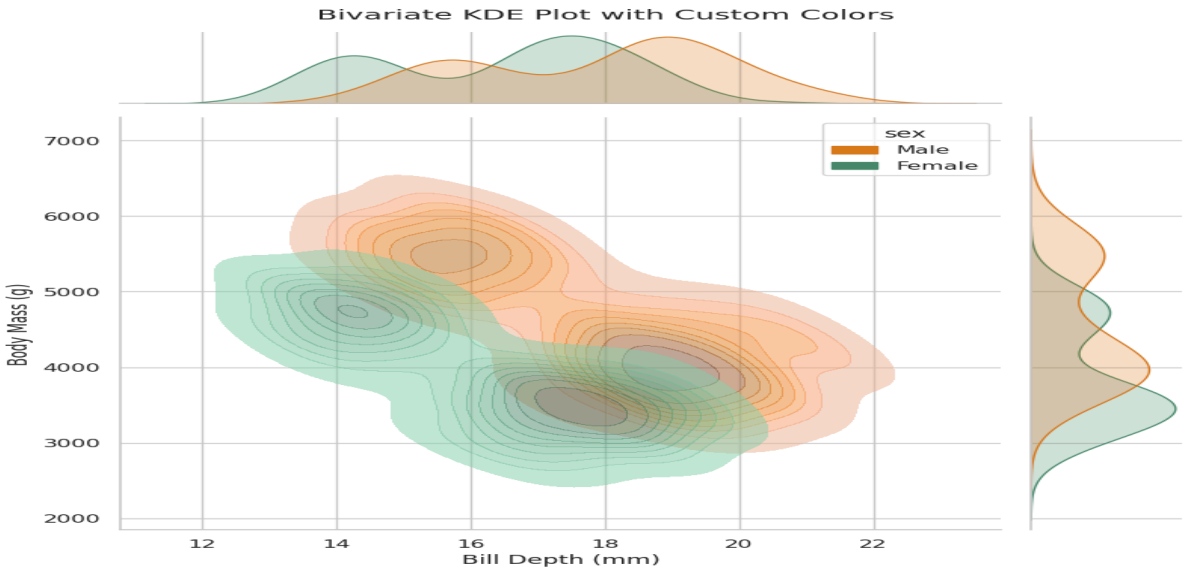
The maximum male bill length and body mass is higher than the female bill length and body mass. Most of the female bill lengths are around 36mm and 46mm whereas most of the male bill lengths are around 42mm and mostly around 50mm. Most of the male have the body mass around 5500 grams and mostly around 4000 grams whereas most of the females have the body mass around 4800 grams and mostly around 3300 grams.

KDE analysis between bill_depth and body_mass based on the species



The Adelie and the Chinstrap have a similar range of bill depth and body mass. Most of them have a bill depth of 18mm with a body mass range between 3100 grams to 4000 grams. But Gentoo, on the other hand, has lower bill depth but has a higher body mass. Most of them have 14mm bill depth and around 4500 grams of body mass.

KDE analysis between bill_depth and body_mass based on the sex



Both male and female, when the bill depth increases, the body mass decreases simultaneously. However, most of the females have a body mass range between 3100 grams to 3900 grams with a bill depth of around 18mm. On the other hand, the males have slightly higher bill depth (around 19mm) with the body mass of approximately 4000 grams.